Learning from Unlabeled Data for Aβ Plaque Type Classification

A Comparative Study of Supervised, Semi-Supervised, and Self-Supervised Learning on Segmented Brain Pathology Crops

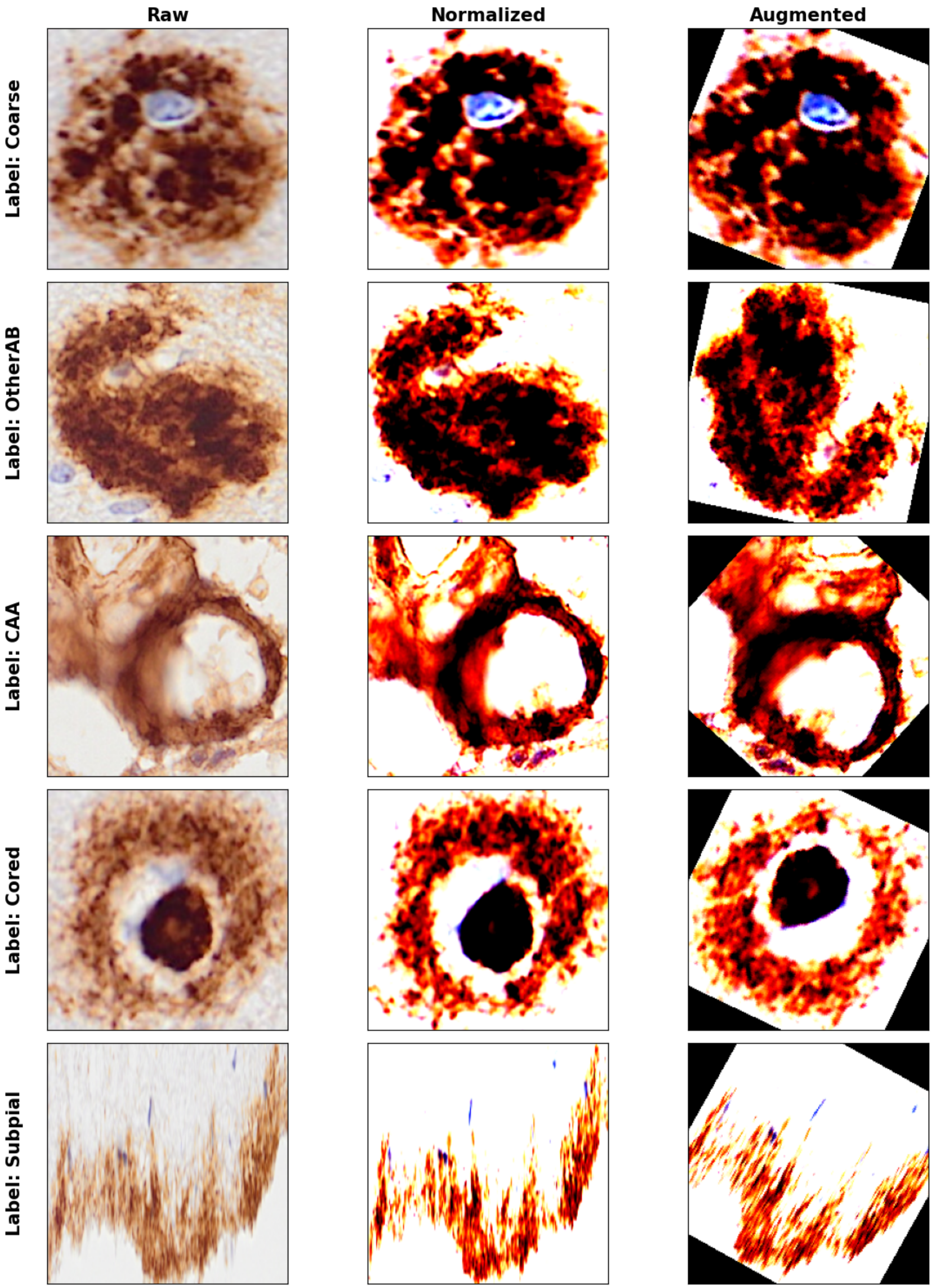
Introduction

- Aim of the project: Improving the Aβ Plaque classification performance
- Prior work by Moreira-Kanaley relied on only annotated plaque images which was expensive to obtain
- Research gap: Missing on approximately 4.7 million unannotated plaque segmented images
- Solution: Utilize a subset of these annotated plaques in the training process using **Semi/Self supervised learning paradigms**

Data Processing

- Labeled plaques and a subset of 10000 unlabeled plaques are extracted from HDF5 files and saved in png image files
- Images are
  - resized to 224*224
  - Normalized
  - Augmented through flip and rotate transformations

Annotated plaques ($n = 589$)	
Class	**Count (%)**
Diffuse	81 (13.7%)
Cored	107 (18.1%)
Compact	63 (10.6%)
Coarse	79 (13.4%)
CAA	74 (12.5%)
Subpial	73 (12.3%)
OtherAB	31 (5.2%)
UndefAB	43 (7.3%)
NonAB	38 (6.4%)
Unlabeled plaques ($n = 4,746,466$)	
Sampled	10,000
Total dataset size	**10,589**



Models

- ResNet18 backbone followed by a projection layer is used across all the methods as the feature extractor.
- A simple linear layer is used for classifier on top of the feature extractor
- Having this shared architecture Supervised, Semi-supervised and Self-supervised methods are implemented
- The models are trained in two ResNet18 weight initialization settings
    - 1) Random weight
    - 2) ImageNet pretrained weight

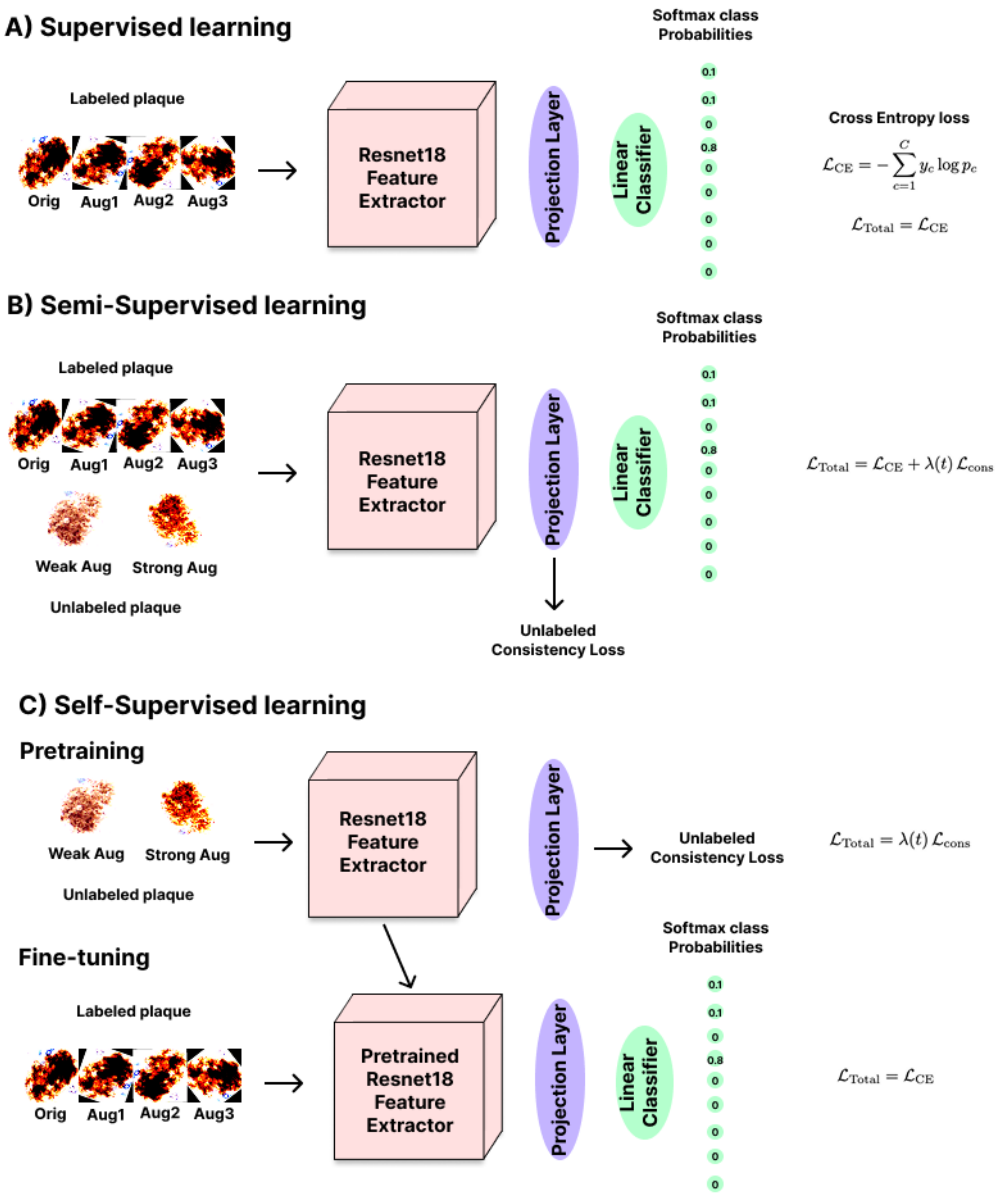
Supervised: only uses labeled plaques and learns classes using Cross entropy loss

Semi-supervised:

- at each training step considers one batch of labeled and one batch of unlabeled samples
- includes a consistency loss between different augmentations of a unlabeled plaque in the total loss optimized

Self-supervised: consists of two training steps

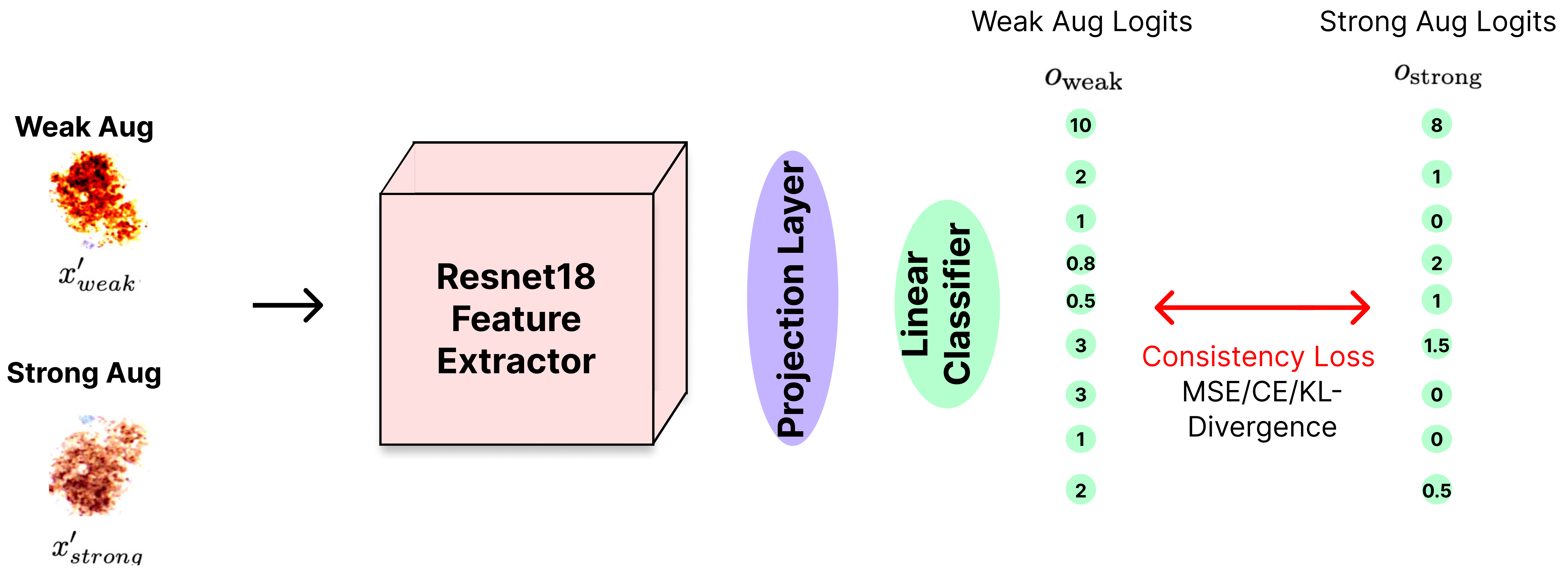
- 1- Pretraining on only the unlabeled samples
- 2- Fine-tuning the pretrained feature extractor on the few labeled plaques



Semi-supervised) Pi-model

Pi-Model

x': Unlabeled Image
o: Logits
p: Predicted Probability
p = SoftMax(o)

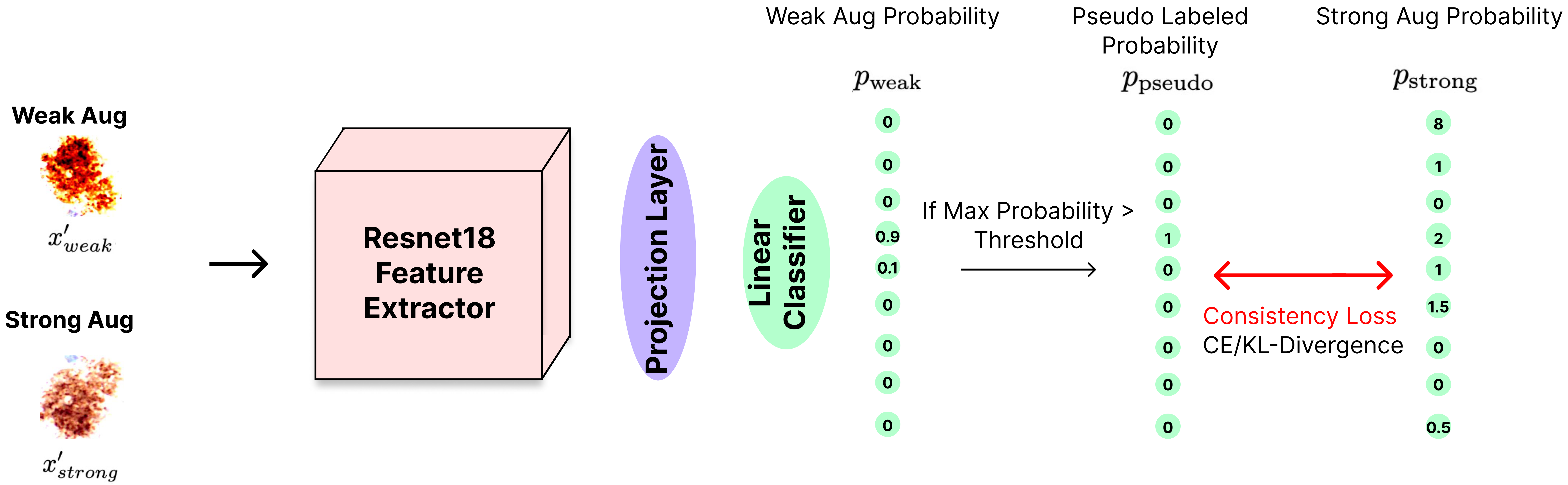


$$
\mathcal{L}_{\text{consistency}}(x'_{\text{weak}}, x'_{\text{strong}}) = \begin{cases} \text{MSE}\left(o_{\text{strong}},\ o_{\text{weak}}\right) = \frac{1}{9}\sum_{c=1}^{9}\left(o_{\text{strong},c} - o_{\text{weak},c}\right)^2 \\ \text{KL}\left(p_{\text{strong}} \,\|\, p_{\text{weak}}\right) = \sum_{c=1}^{9} p_{\text{strong},c} \log \frac{p_{\text{strong},c}}{p_{\text{weak},c}} \\ \text{CE}\left(p_{\text{strong}},\ p_{\text{weak}}\right) = -\sum_{c=1}^{9} p_{\text{strong},c} \log p_{\text{weak},c} \end{cases}
$$

Semi-supervised) FixMatch

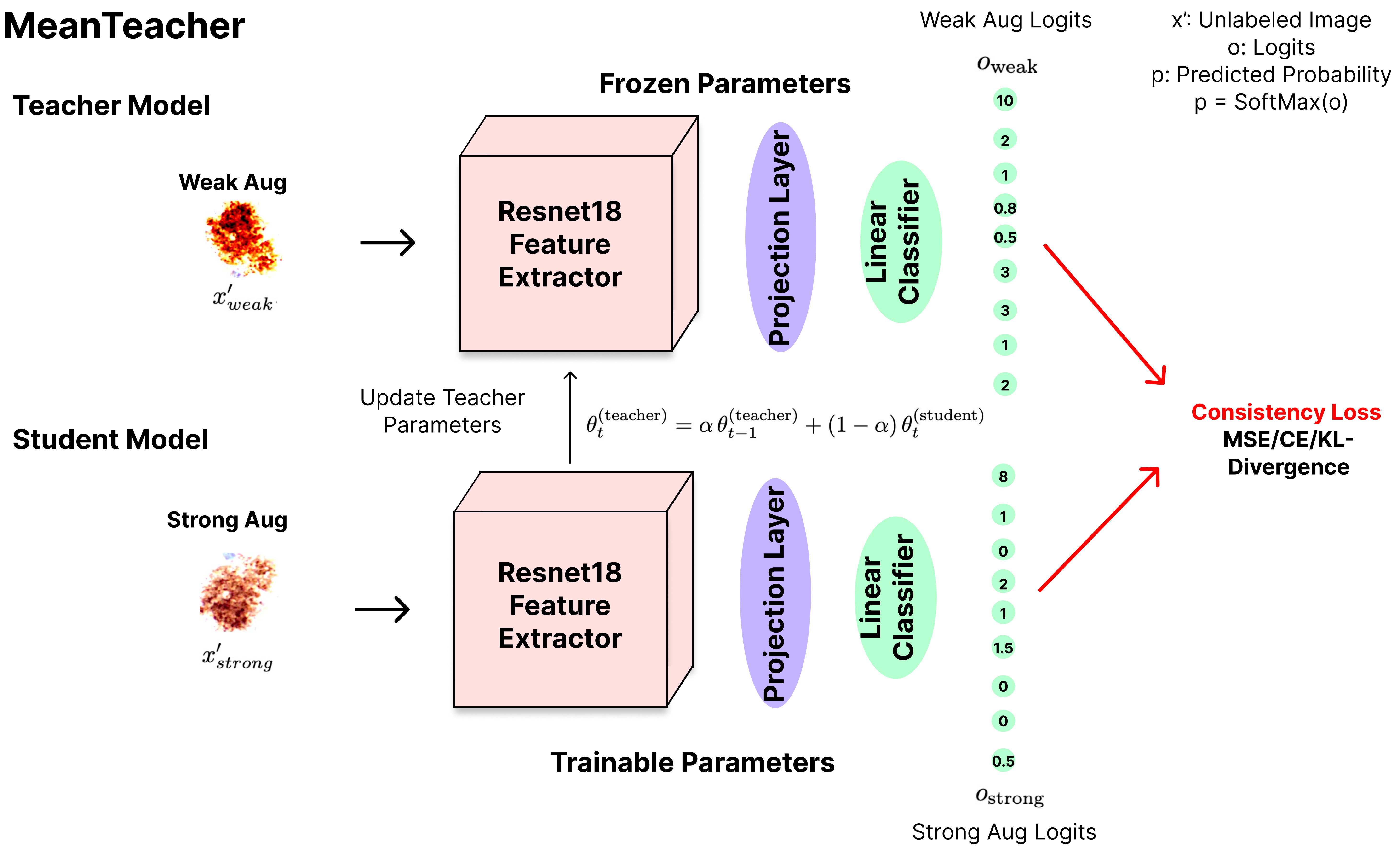
FixMatch

x': Unlabeled Image
p: Predicted Softmax Class Probability



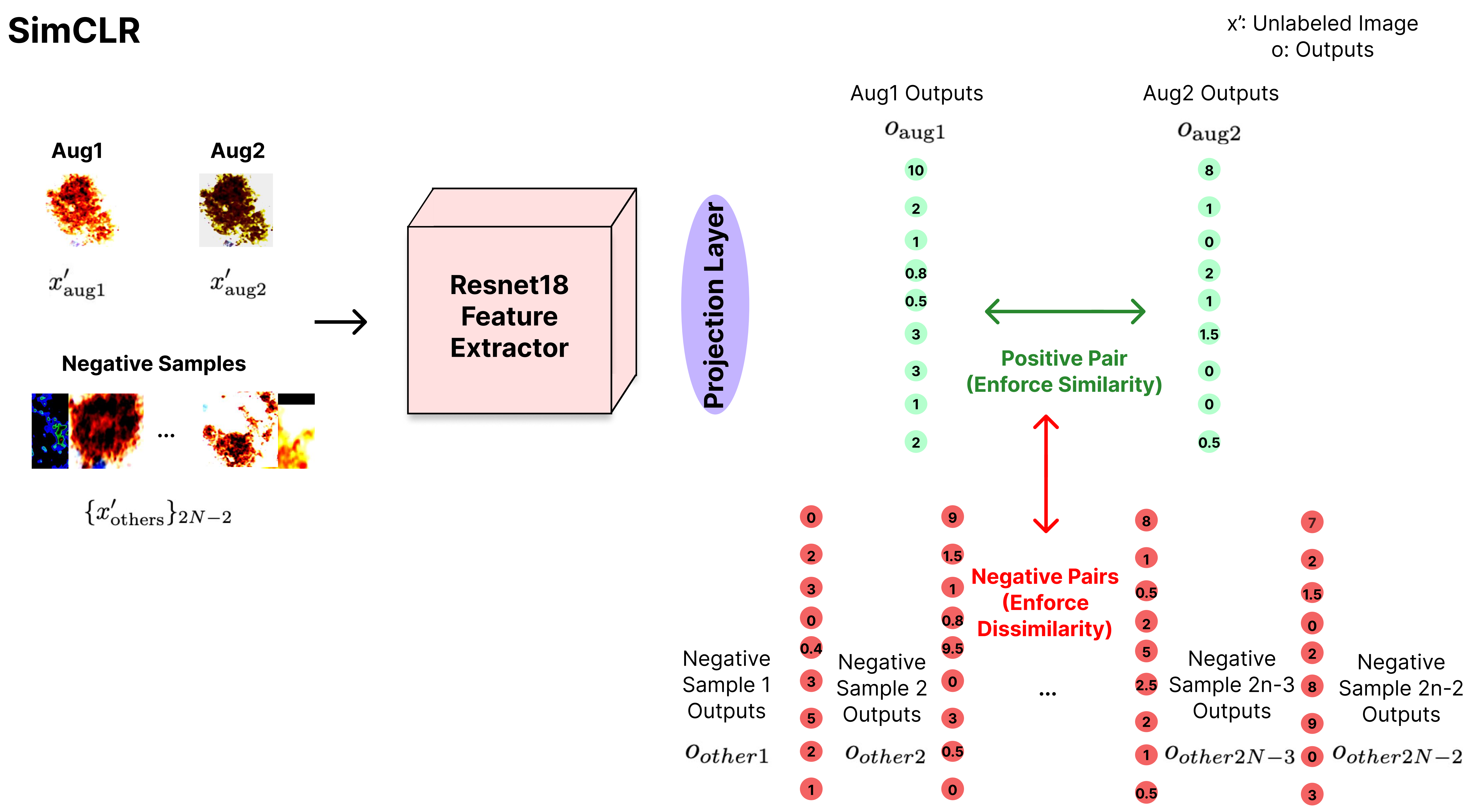
$$\mathcal{L}_{\text{consistency}}(x'_{\text{weak}}, x'_{\text{strong}}) = \begin{cases} \mathrm{KL}\left(p_{\text{strong}} \,\|\, p_{\text{pseudo}}\right) = \sum_{c=1}^{9} p_{\text{strong},c} \log \frac{p_{\text{strong},c}}{p_{\text{pseudo},c}} \\ \mathrm{CE}\left(p_{\text{strong}},\ p_{\text{pseudo}}\right) = -\sum_{c=1}^{9} p_{\text{strong},c} \log p_{\text{pseudo},c} \end{cases}$$

Semi-supervised) MeanTeacher



$$\mathcal{L}_{\text{consistency}}(x'_{\text{weak}}, x'_{\text{strong}}) = \begin{cases} \text{MSE}\left(o_{\text{strong}},\ o_{\text{weak}}\right) = \frac{1}{9}\sum_{c=1}^{9}\left(o_{\text{strong},c} - o_{\text{weak},c}\right)^2 \\ \text{KL}\left(p_{\text{strong}} \,\|\, p_{\text{weak}}\right) = \sum_{c=1}^{9} p_{\text{strong},c} \log \frac{p_{\text{strong},c}}{p_{\text{weak},c}} \\ \text{CE}\left(p_{\text{strong}},\ p_{\text{weak}}\right) = -\sum_{c=1}^{9} p_{\text{strong},c} \log p_{\text{weak},c} \end{cases}$$

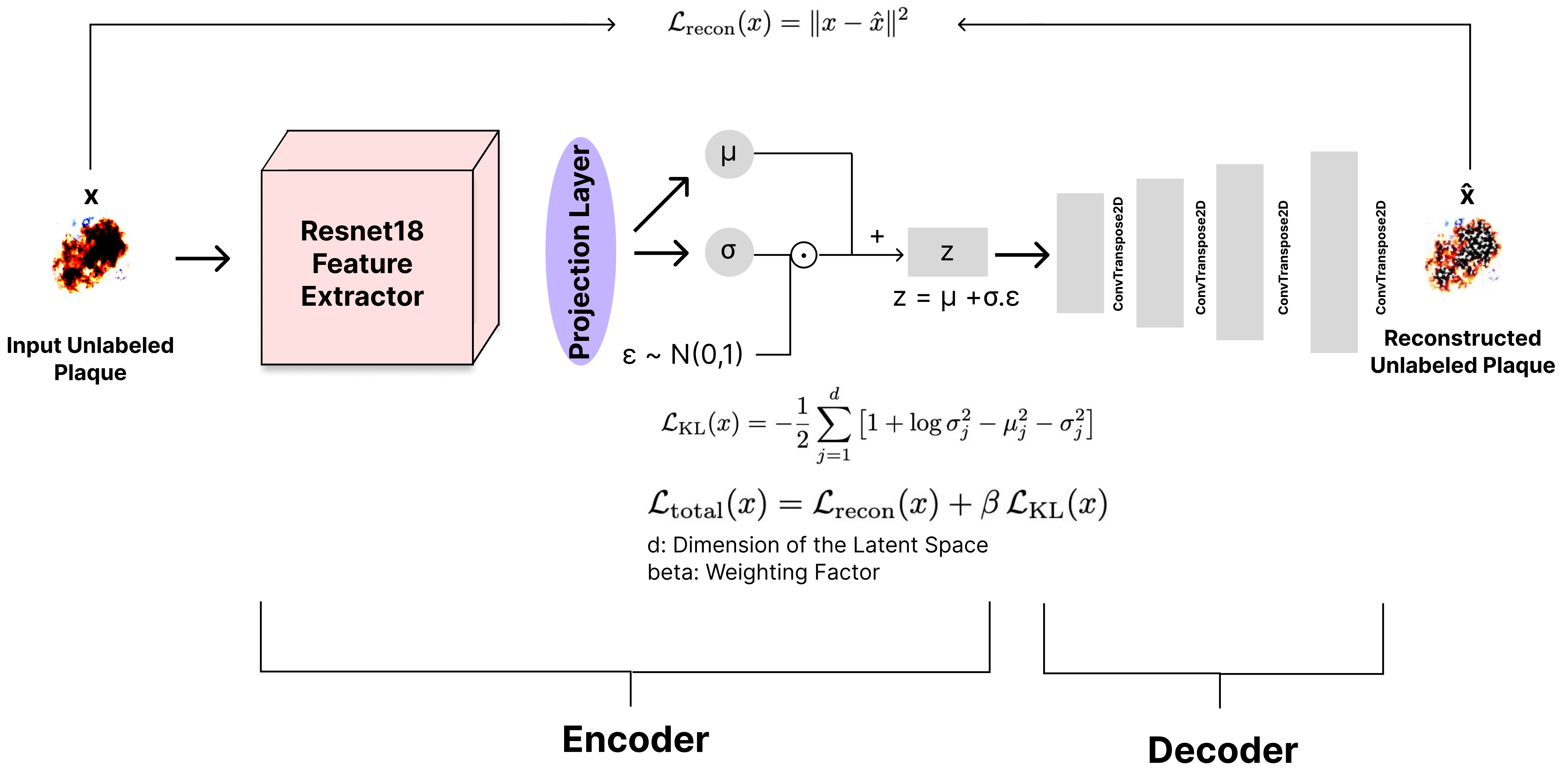
Self-supervised) SimCLR



$$\mathcal{L}_{\text{contrast}}(x'_{\text{aug1}}, x'_{\text{aug2}}, \{x'_{\text{others}}\}_{2N-2}) = -\log \frac{\exp(\text{sim}(o_{\text{aug1}}, o_{\text{aug2}})/\tau)}{\sum_{o\in\{o_{\text{aug2}}\}\cup\{o_{\text{others}}\}} \exp(\text{sim}(o_{\text{aug1}}, o)/\tau)}$$

Self-supervised)
Variational Auto Encoder (VAE)

VAE

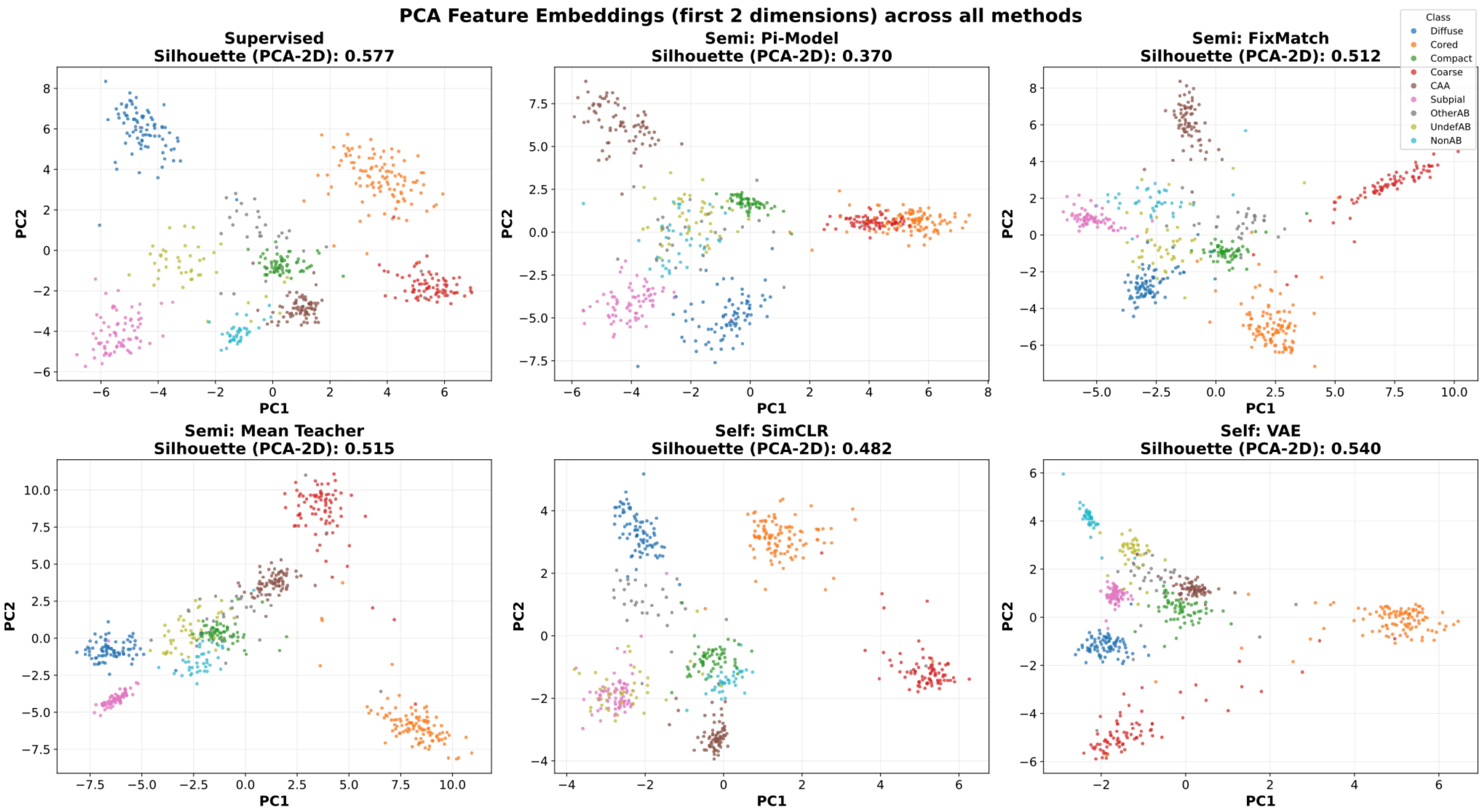


Hyperparameter tuning

- There are various hyper parameters in this project
    - Runtime hyperparameters like batch size, learning rate, weight decay, …
    - Architecture hyper parameters like projection layer size, drop out rate, …
    - Method specific hyper parameters like temperature in SimCLR
- We perform a round of hyper parameter tuning using Optuna's Tree-structured Parzen Estimator (TPE) by running 20 trials (20 combinations of hyper parameters overall)
- This hyperparameter tuning is performed for each model and the best hyperparmeters are selected based on 5-fold cross validation macro f1-score of the model

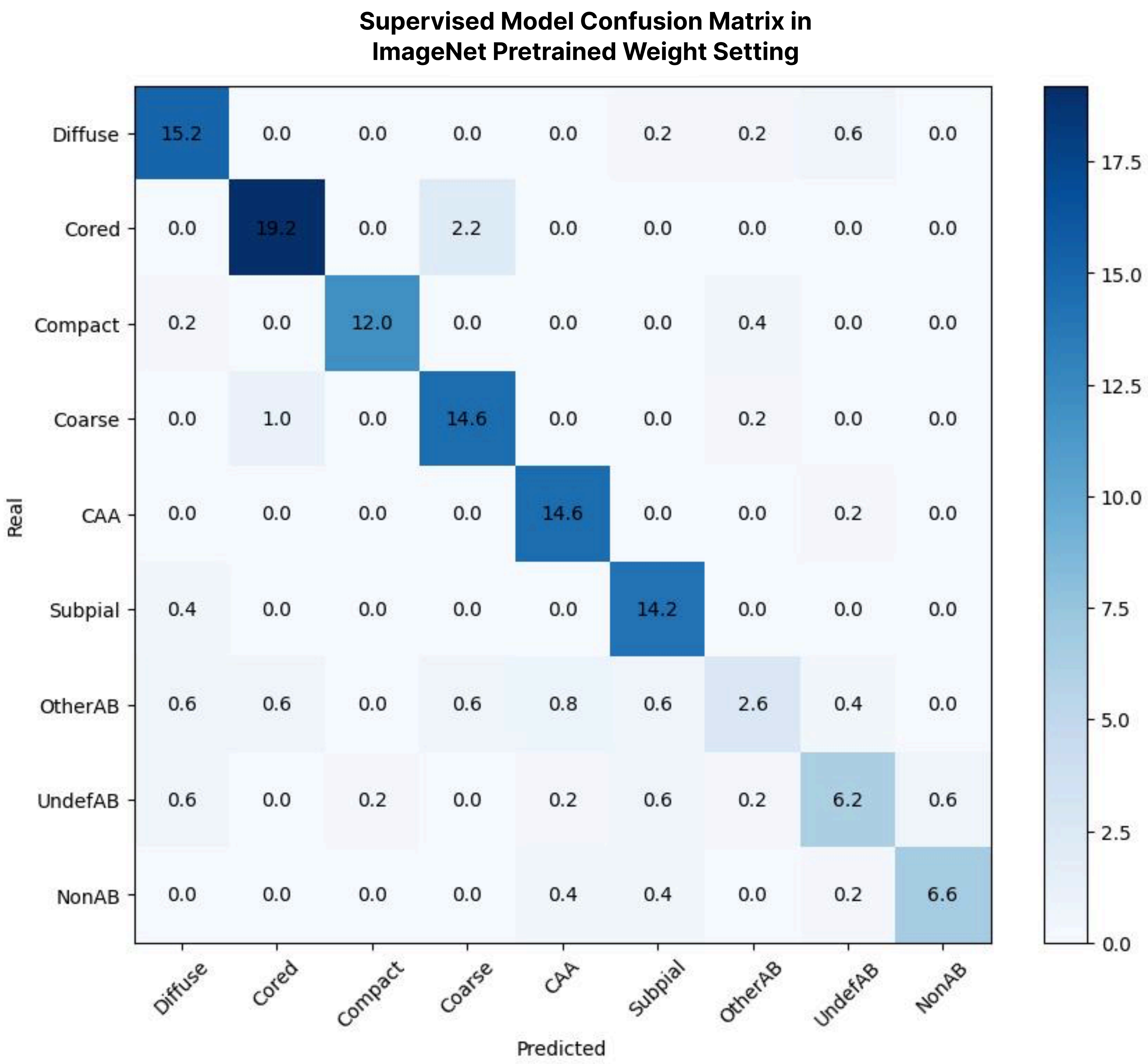
Results

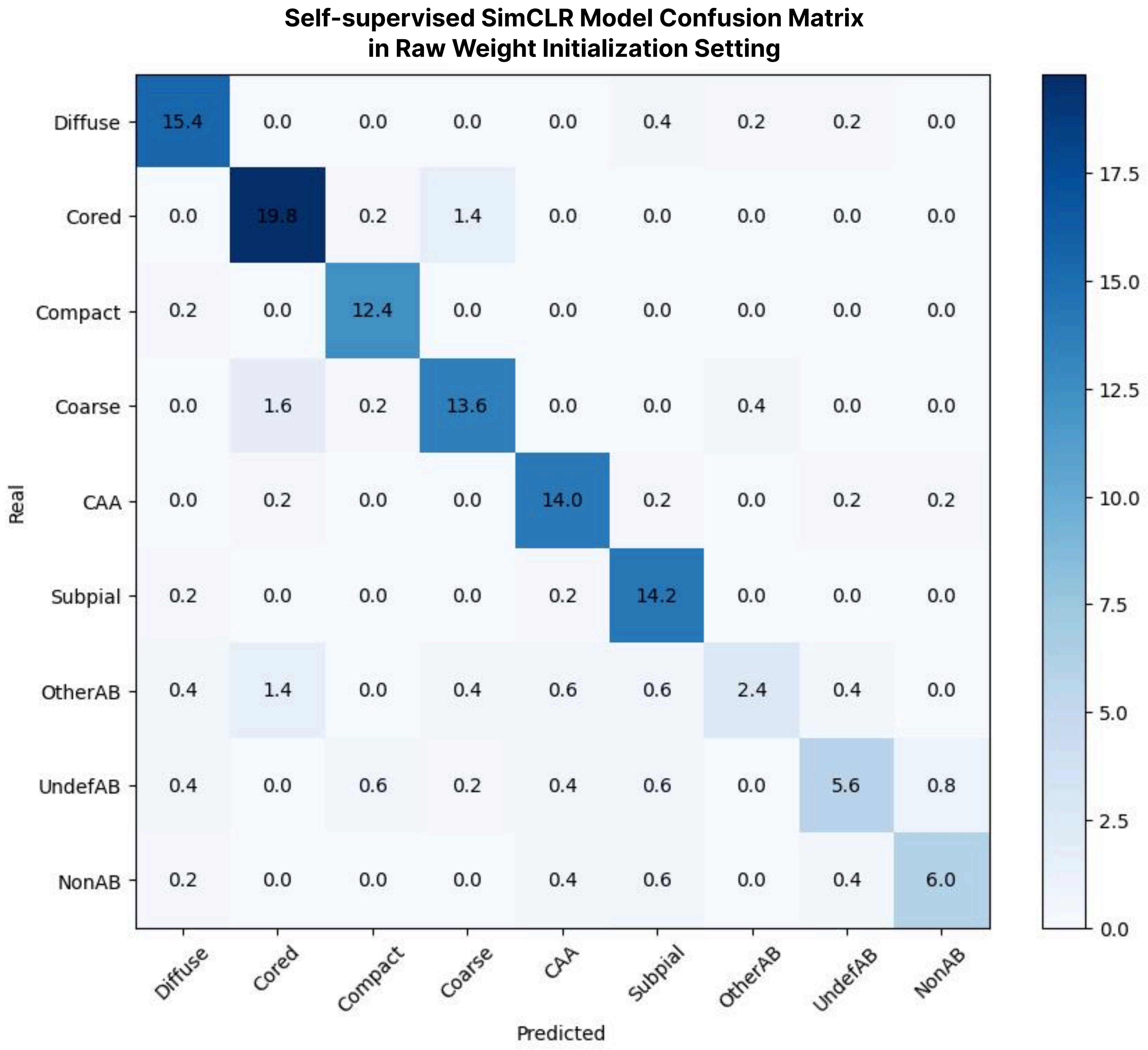
Learning paradigm	Method	Macro-F1 score: Random initialization	Macro-F1 score: ImageNet pretrained
Supervised	Supervised baseline	0.714 ± 0.034	**0.852 ± 0.033**
Semi-supervised	Pi-Model	0.715 ± 0.036	0.806 ± 0.034
Semi-supervised	FixMatch	0.725 ± 0.061	0.801 ± 0.016
Semi-supervised	Mean Teacher	0.725 ± 0.035	0.832 ± 0.017
Self-supervised	SimCLR pretraining + fine-tuning	**0.750 ± 0.046**	0.835 ± 0.054
Self-supervised	VAE pretraining + fine-tuning	0.735 ± 0.037	0.834 ± 0.047



Model	Silhouette (PCA 2D)	Silhouette (all dims)	Feature Dim
Supervised	**0.577**	0.613	64
Semi: Pi-Model	0.370	0.578	64
Semi: Mean Teacher	0.515	0.662	128
Semi: FixMatch	0.512	0.642	64
Self: VAE	0.540	**0.700**	32
Self: SimCLR	0.482	0.690	32

	Supervised (ImageNet)			SimCLR (random initialization)			
Class	**P**	**R**	**F1**	**P**	**R**	**F1**	**Support**
Diffuse	0.892 ± 0.032	0.938 ± 0.108	0.912 ± 0.067	0.857 ± 0.110	0.878 ± 0.071	0.862 ± 0.054	16.2 ± 0.447
Cored	0.927 ± 0.048	0.898 ± 0.074	0.910 ± 0.038	0.838 ± 0.037	0.823 ± 0.104	0.828 ± 0.061	21.4 ± 0.548
Compact	0.985 ± 0.034	0.953 ± 0.043	0.967 ± 0.018	0.906 ± 0.120	0.937 ± 0.101	0.916 ± 0.085	12.6 ± 0.548
Coarse	0.845 ± 0.074	0.924 ± 0.032	0.881 ± 0.043	0.836 ± 0.073	0.898 ± 0.105	0.860 ± 0.035	15.8 ± 0.447
CAA	0.916 ± 0.064	0.987 ± 0.030	0.949 ± 0.042	0.765 ± 0.068	0.852 ± 0.086	0.802 ± 0.047	14.8 ± 0.447
Subpial	0.893 ± 0.081	0.972 ± 0.038	0.929 ± 0.047	0.870 ± 0.050	0.929 ± 0.072	0.899 ± 0.057	14.6 ± 0.548
OtherAB	0.617 ± 0.371	0.419 ± 0.363	**0.469 ± 0.312**	0.467 ± 0.361	0.224 ± 0.142	**0.297 ± 0.194**	6.2 ± 0.447
UndefAB	0.818 ± 0.172	0.717 ± 0.115	0.763 ± 0.138	0.628 ± 0.104	0.533 ± 0.165	0.573 ± 0.135	8.6 ± 0.548
NonAB	0.921 ± 0.072	0.871 ± 0.089	0.891 ± 0.039	0.755 ± 0.053	0.711 ± 0.204	0.717 ± 0.127	7.6 ± 0.548
Macro avg	0.868 ± 0.040	0.853 ± 0.034	0.852 ± 0.033	0.769 ± 0.061	0.754 ± 0.037	0.750 ± 0.046	117.8 ± 0.447
Weighted avg	0.889 ± 0.025	0.893 ± 0.019	0.885 ± 0.022	0.804 ± 0.037	0.810 ± 0.027	0.798 ± 0.031	117.8 ± 0.447


Supervised Model Confusion Matrix in
ImageNet Pretrained Weight Setting
Real
Predicted
Diffuse
Cored
Compact
Coarse
CAA
Subpial
OtherAB
UndefAB
NonAB
15.2 0.0 0.0 0.0 0.0 0.2 0.2 0.6 0.0
0.0 19.2 0.0 2.2 0.0 0.0 0.0 0.0 0.0
0.2 0.0 12.0 0.0 0.0 0.0 0.4 0.0 0.0
0.0 1.0 0.0 14.6 0.0 0.0 0.2 0.0 0.0
0.0 0.0 0.0 0.0 14.6 0.0 0.0 0.2 0.0
0.4 0.0 0.0 0.0 0.0 14.2 0.0 0.0 0.0
0.6 0.6 0.0 0.6 0.8 0.6 2.6 0.4 0.0
0.6 0.0 0.2 0.0 0.2 0.6 0.2 6.2 0.6
0.0 0.0 0.0 0.0 0.4 0.4 0.0 0.2 6.6
17.5
15.0
12.5
10.0
7.5
5.0
2.5
0.0


Self-supervised SimCLR Model Confusion Matrix
in Raw Weight Initialization Setting
Real
Predicted
Diffuse
Cored
Compact
Coarse
CAA
Subpial
OtherAB
UndefAB
NonAB
15.4 0.0 0.0 0.0 0.0 0.4 0.2 0.2 0.0
0.0 19.8 0.2 1.4 0.0 0.0 0.0 0.0 0.0
0.2 0.0 12.4 0.0 0.0 0.0 0.0 0.0 0.0
0.0 1.6 0.2 13.6 0.0 0.0 0.4 0.0 0.0
0.0 0.2 0.0 0.0 14.0 0.2 0.0 0.2 0.2
0.2 0.0 0.0 0.0 0.2 14.2 0.0 0.0 0.0
0.4 1.4 0.0 0.4 0.6 0.6 2.4 0.4 0.0
0.4 0.0 0.6 0.2 0.4 0.6 0.0 5.6 0.8
0.2 0.0 0.0 0.0 0.4 0.6 0.0 0.4 6.0
17.5
15.0
12.5
10.0
7.5
5.0
2.5
0.0

Results Comparison

- Previous work by Moreira-Kanaley achieved a macro f1-score of 0.77 using ensemble of ResNet50 models and supervised learning
- But our best setting got macro f1-score of 0.85 although we used ResNet18 which is much smaller model
- Explanation:
  - We had access to 589 annotated plaques whereas previous work reported 315 annotated plaques (likely because of new annotations added)
  - Segmented images are resized to 224*224 in this study as opposed to 128*128 which was done in the previous work
  - Training and evaluation protocols difference might have also caused this performance improvement

Conclusion

- Unlabeled objectives provide limited downstream macro-F1 improvement when the backbone is already initialized with strong ImageNet-pretrained weights
- Unlabeled data become significantly more valuable when strong pretrained representations are not available
- In Semi/Self supervised learning the overall geometry of the features learnt is more clustered than the supervised learning (due to higher silhouette score)
- However these improved representation geometry does not necessarily translate into improved downstream classification performance.

Limitations

- Only considered ResNet18 model (due to resource constraints specially in the hyperparmaeter tuning part)
- A small subset of available unlabeled samples was used (10,000 out of 4.7 million)
- Didn’t consider optimizing data augmentation techniques
- The hyperparamter search was limited to only 20 trials (20 different combinations of hyper parameters)
- Class imbalance especially for non-primary plaques